

Zhifeng Gu

zhifeng.gu@connect.polyu.hk | <https://neal2020github.github.io>

Education

The Hong Kong Polytechnic University , Hong Kong	Sep 2023 – Present
• Doctor of Philosophy, Advised by Prof. Bing Wang • Research Interests: Spatial Intelligence, Embodied Agents, Embodied RSI	
Beihang University , Beijing, China	Sep 2019 – Jun 2023
• Bachelor of Software Engineering, Advised by Prof. Lu Sheng	GPA: 3.8/4.0

Internships

XPENG Robotics , Shenzhen, China	Apr 2026 – Present
• Research Direction: Recursive Self-Improvement in Embodied Intelligence	

Publications

Done, But Not Sure: Disentangling World Completion from Self-Termination in Embodied Agents

Ying Chen, Lihuang Fang, Rui Jiang, Mingxu Wang, Zhifeng Gu, Lei Yi, Jie Chen
NeurIPS 2026

R³L: Reasoning 3D Layouts from Relative Spatial Relations

Zhifeng Gu^{*}, Yuqi Wang^{*}, Bing Wang
ICML 2026

MMOne: Representing Multiple Modalities in One Scene

Zhifeng Gu, Bing Wang
ICCV 2025

Research Experience

Spatial Reasoning for 3D Layout Generation	Jul 2025 – Jan 2026
SPACE Group, HK PolyU, Advised by Prof. Bing Wang	
• Present a general framework for 3D layout generation, featuring invariant spatial decomposition and consistent spatial imagination to improve the reliability and consistency of multi-hop relative spatial reasoning.	
Multimodal Scene Representation	Sep 2024 – Mar 2025
SPACE Group, HK PolyU, Advised by Prof. Bing Wang	
• Propose a general framework for representing multiple modalities in one scene, with a modality modeling module and a multimodal decomposition mechanism to address modality conflicts.	

Selected Honors and Awards

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- Outstanding Graduate of Beihang University

Selected Skills

Python, PyTorch, C/C++ , CUDA, Blender, etc.